程序控制

Process Control

工程數學工具箱

Engineering Mathematics Toolbox

中英文對照版 Bilingual Edition

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工程數學工具箱

Engineering Mathematics Toolbox

本工具箱涵蓋 Chapter 1–17 使用到的數學先備知識。

This toolbox covers the mathematical prerequisites for Chapters 1–17.

M1. 分部積分 Integration by Parts

公式：$\int u\,dv = uv - \int v\,du$

Formula: $\int u\,dv = uv - \int v\,du$

LIATE 選擇法則（選 u 時，由左至右優先）： 對數 > 反三角 > 代數 > 三角 > 指數

LIATE Rule *(priority for choosing u, left to right): Logarithmic > Inverse trig > Algebraic > Trigonometric > Exponential*

直覺理解：分部積分是「乘積法則的逆運算」。

Intuition: *Integration by parts is the product rule run in reverse.*

範例：推導 $L\{t\} = 1/s^2$

Example: *Derive* $L\{t\} = 1/s^2$

Step 1: $L\{t\} = \int_0^\infty t\cdot e^{-st}\,dt$

Step 2: $u = t,\ dv = e^{-st}\,dt \rightarrow du = dt,\ v = -(1/s)e^{-st}$

Step 3: $= [t\cdot(-1/s)e^{-st}]_0^\infty + (1/s)\int_0^\infty e^{-st}\,dt$

Step 4: 邊界項 → 0（羅必達），$= 0 + (1/s)\cdot(1/s) = 1/s^2$

Step 4: *Boundary term → 0 (L'Hôpital),* $= 0 + (1/s)\cdot(1/s) = 1/s^2$

練習題：計算 $L\{t\cdot e^{-2t}\}$。

Practice: *Compute* $L\{t\cdot e^{-2t}\}$.

Step 1: $u = t,\ dv = e^{-(s+2)t}\,dt \rightarrow du = dt,\ v = -1/(s+2)\cdot e^{-(s+2)t}$

Step 2: $= [-t/(s+2)\cdot e^{-(s+2)t}]_0^\infty + 1/(s+2)\cdot\int_0^\infty e^{-(s+2)t}\,dt$

✓ $= 0 + 1/(s+2)\cdot 1/(s+2) = 1/(s+2)^2$

課程連結：Ch1-2（Unit 6）推導 $L\{t\}$、$L\{t^n\}$、微分性質 $L\{f'\}=sF(s)-f(0)$ 時皆使用分部積分。Ch6（高階系統推導）也用到。

Connection: *Ch1-2 (Unit 6) for deriving* $L\{t\}$, $L\{t^n\}$, *and derivative property* $L\{f'\}=sF(s)-f(0)$. *Also Ch6 (higher-order derivations).*

M2. 瑕積分 Improper Integrals

定義：$\int_0^\infty f(t)\,dt = \lim_{b\to\infty} \int_0^b f(t)\,dt$

Definition: $\int_0^\infty f(t)\,dt = \lim_{b\to\infty} \int_0^b f(t)\,dt$

收斂：極限存在且有限；發散：否則。

Convergence: *Limit exists and is finite; diverges otherwise.*

直覺理解：e^{-st} 在 Re(s)>0 時保證積分收斂。

Intuition: *The e^{-st} factor ensures convergence when Re(s)>0.*

範例：$\int_0^\infty e^{-st}\,dt = 1/s$ (s>0)

Example: $\int_0^\infty e^{-st}\,dt = 1/s$ *(s>0)*

Step: $\lim(b\to\infty)\,[-(1/s)e^{-st}]_0^b = 0 + 1/s = 1/s = L\{1\}$

練習題：計算 $\int_0^\infty 3e^{-2t}\,dt$。

Practice: *Evaluate* $\int_0^\infty 3e^{-2t}\,dt$.

Step 1: $= 3\cdot\lim(b\to\infty)\,[-1/2\cdot e^{-2t}]_0^b = 3\cdot(0-(-1/2))$

✓ $= 3/2$

課程連結：Ch1-2（Unit 6）Laplace 轉換 $F(s)=\int_0^\infty f(t)e^{-st}dt$ 本身就是瑕積分，Re(s)>0 為收斂條件，貫穿 Ch3–Ch8。

Connection: *The Laplace transform is itself an improper integral. The Re(s)>0 convergence condition recurs throughout Ch3–Ch8.*

M3. 羅必達法則 L'Hôpital's Rule

法則：若 lim f(x)/g(x) 為 0/0 或 ∞/∞，則 lim f(x)/g(x) = lim f'(x)/g'(x)

Rule: *If lim f(x)/g(x) is 0/0 or ∞/∞, then lim f(x)/g(x) = lim f'(x)/g'(x)*

範例：$\lim(t\to\infty)\,t\cdot e^{-st} = \lim(t\to\infty)\,t/e^{st} = \lim(t\to\infty)\,1/(s\cdot e^{st}) = 0$

Example: $\lim(t\to\infty)\,t\cdot e^{-st} = \lim(t\to\infty)\,t/e^{st} = \lim(t\to\infty)\,1/(s\cdot e^{st}) = 0$

練習題：用 FVT 計算 $\lim(s\to 0)\,s\cdot 5/(s(s+2))$。

Practice: *Compute* $\lim(s\to 0)\,s\cdot 5/(s(s+2))$ *via FVT.*

Step 1: $= \lim(s\to 0)\,5/(s+2)$

✓ $= 5/2$

課程連結：Ch1-2（Unit 6）推導 $L\{t\}=1/s^2$ 的邊界項為零之證明。Ch3（重根極限）、Ch6（串聯系統）也需要。

Connection: *Ch1-2 (Unit 6) boundary term proof for* $L\{t\}=1/s^2$. *Also Ch3 (repeated poles) and Ch6 (cascaded systems).*

M4. 歐拉公式 Euler's Formula

公式：$e^{j\theta} = \cos\theta + j\sin\theta$

Formula: $e^{j\theta} = \cos\theta + j\sin\theta$

推論：$\sin\theta = (e^{j\theta} - e^{-j\theta})/(2j)$ $\quad$ $\cos\theta = (e^{j\theta} + e^{-j\theta})/2$

Corollaries: $\sin\theta = (e^{j\theta} - e^{-j\theta})/(2j)$ $\quad$ $\cos\theta = (e^{j\theta} + e^{-j\theta})/2$

來源（Taylor 展開）：$e^{j\theta} = 1 + j\theta + (j\theta)^2/2! + \ldots \rightarrow$ 實部 $= \cos\theta$，虛部 $= \sin\theta$

Origin (Taylor expansion): *$e^{j\theta} = 1 + j\theta + (j\theta)^2/2! + \ldots \rightarrow$ real part $= \cos\theta$, imaginary part $= \sin\theta$*

範例：$L\{\cos(kt)\} = \mathrm{Re}[1/(s-jk)] = s/(s^2+k^2)$

Example: *$L\{\cos(kt)\} = \mathrm{Re}[1/(s-jk)] = s/(s^2+k^2)$*

練習題：將 e^(jπ/3) 化為直角座標 a + jb。

Practice: *Express e^(jπ/3) in rectangular form a + jb.*

Step 1: e^(jπ/3) = cos(π/3) + j sin(π/3)

✓ $= 1/2 + j\sqrt{3}/2$

課程連結：Ch1-2（Unit 6）推導 L{sin(kt)} 和 L{cos(kt)}。Ch7（欠阻尼響應）、Ch15（s=jω 頻率響應）、Ch16（Bode 穩定性）。

Connection: *Ch1-2 (Unit 6) for L{sin(kt)} and L{cos(kt)}. Also Ch7 (underdamped), Ch15 (frequency response), Ch16 (Bode).*

M5. 複數基礎 Complex Number Basics

定義：$j = \sqrt{(-1)}$, $j^2 = -1$；$z = a + jb$；共軛 $\bar{z} = a - jb$；模 $|z| = \sqrt{(a^2+b^2)}$

Definition: *$j = \sqrt{(-1)}$, $j^2 = -1$; $z = a + jb$; conjugate $\bar{z} = a - jb$; modulus $|z| = \sqrt{(a^2+b^2)}$*

關鍵恆等式：$(s - jk)(s + jk) = s^2 + k^2$ — 消去虛部、產生實數分母

Key identity: *$(s - jk)(s + jk) = s^2 + k^2$ — eliminates imaginary parts, produces real denominator*

範例：$1/(s-3j) \times (s+3j)/(s+3j) = (s+3j)/(s^2+9)$

Example: *$1/(s-3j) \times (s+3j)/(s+3j) = (s+3j)/(s^2+9)$*

練習題：已知 G(s) = 1/(s+2)，求 |G(j3)| 和 ∠G(j3)。

Practice: *Given G(s) = 1/(s+2), find |G(j3)| and ∠G(j3).*

Step 1: G(j3) = 1/(2+j3), $|G| = 1/\sqrt{(4+9)} = 1/\sqrt{13}$

✓ $\angle G = -\arctan(3/2) \approx -56.3°$

課程連結：Ch1-2（Unit 6 & 8）。Ch3（共軛根）、Ch7（阻尼比）、Ch8（特徵方程式）、Ch13（極點位置）、Ch15-16（頻率響應）、Ch17（根軌跡）皆需複數運算。

Connection: *Ch1-2 (Unit 6 & 8). Complex arithmetic is essential in Ch3, Ch7, Ch8, Ch13, Ch15-16 (frequency response), and Ch17 (root locus).*

M6. 雙曲函數 Hyperbolic Functions

定義：$\sinh(x) = (e^x - e^{-x})/2$ $\quad$ $\cosh(x) = (e^x + e^{-x})/2$

Definition: *$\sinh(x) = (e^x - e^{-x})/2$ $\cosh(x) = (e^x + e^{-x})/2$*

恆等式：$\cosh^2(x) - \sinh^2(x) = 1$（對照 $\cos^2+\sin^2=1$）

Identity: *$\cosh^2(x) - \sinh^2(x) = 1$ (compare $\cos^2+\sin^2=1$)*

對照：$L\{\sin(kt)\} = k/(s^2+k^2) \leftrightarrow L\{\sinh(kt)\} = k/(s^2-k^2)$ 穩定振盪 ↔ 不穩定增長（$+k^2 \leftrightarrow -k^2$）

Comparison: *$L\{\sin(kt)\} = k/(s^2+k^2) \leftrightarrow L\{\sinh(kt)\} = k/(s^2-k^2)$ Stable oscillation ↔ unstable growth ($+k^2 \leftrightarrow -k^2$)*

練習題：驗證 $L\{\sinh(3t)\} = 3/(s^2-9)$。

Practice: *Verify that $L\{\sinh(3t)\} = 3/(s^2-9)$.*

Step 1: $\sinh(3t) = (e^{3t} - e^{-3t})/2$

Step 2: $L = (1/2)[1/(s-3) - 1/(s+3)] = (1/2)\cdot 6/(s^2-9)$

✓ $= 3/(s^2-9)$ ✓

課程連結：Ch1-2（Unit 6 Laplace 表、Unit 8 正實根不穩定響應）。Ch13（RHP 極點）也涉及不穩定指數響應。

Connection: *Ch1-2 (Unit 6 Laplace table, Unit 8 unstable real-root responses). Also Ch13 (RHP poles).*

M7. 部分分式分解技巧 Partial Fraction Techniques

Case 1 相異實根 — Cover-up 法：$F(s) = 1/[(s+1)(s+2)] \rightarrow A = 1/(-1+2) = 1$, $B = 1/(-2+1) = -1 \rightarrow f(t) = e^{-t} - e^{-2t}$

Case 1 Distinct real roots — Cover-up method: *$F(s) = 1/[(s+1)(s+2)] \rightarrow A = 1/(-1+2) = 1$, $B = 1/(-2+1) = -1 \rightarrow f(t) = e^{-t} - e^{-2t}$*

Case 2 重根 — 微分法：$F(s) = s/(s+1)^2 \rightarrow B_2 = s|_{s=-1} = -1$, $B_1 = d/ds[s]|_{s=-1} = 1 \rightarrow f(t) = e^{-t} - t\cdot e^{-t}$

Case 2 Repeated roots — Differentiation: *$F(s) = s/(s+1)^2 \rightarrow B_2 = s|_{s=-1} = -1$, $B_1 = d/ds[s]|_{s=-1} = 1 \rightarrow f(t) = e^{-t} - t\cdot e^{-t}$*

Case 3 共軛複數根 — 配方法：$F(s) = 1/(s^2+2s+5) = 1/[(s+1)^2+2^2] = (1/2)\cdot 2/[(s+1)^2+2^2] \rightarrow f(t) = (1/2)e^{-t}\sin(2t)$

Case 3 Complex conjugate roots — Completing the square: *$F(s) = 1/(s^2+2s+5) = 1/[(s+1)^2+2^2] = (1/2)\cdot 2/[(s+1)^2+2^2] \rightarrow f(t) = (1/2)e^{-t}\sin(2t)$*

練習題：分解 $5/((s+1)(s+3))$ 為部分分式。

Practice: *Decompose $5/((s+1)(s+3))$ into partial fractions.*

Step 1: $5/((s+1)(s+3)) = A/(s+1) + B/(s+3)$

Step 2: Cover-up: $A = 5/(-1+3) = 5/2$, $B = 5/(-3+1) = -5/2$

✓ $= (5/2)\cdot 1/(s+1) - (5/2)\cdot 1/(s+3)$

課程連結：Ch1-2（Unit 8）反 Laplace 核心工具。Ch3（系統性 PFD 複習）、Ch6（串聯系統）、Ch8（閉迴路響應）、Ch12（案例分析）。

Connection: *Ch1-2 (Unit 8) primary inverse-Laplace tool. Also Ch3 (systematic PFD review), Ch6, Ch8 (closed-loop), Ch12 (case studies).*

M8. 積分因子法推導 Integrating Factor Derivation

問題：$dy/dt + P(t)\cdot y = Q(t) \rightarrow \mu(t) = e^{\int P(t)dt}$ 從何而來？

Problem: *$dy/dt + P(t)\cdot y = Q(t) \rightarrow$ where does $\mu(t) = e^{\int P(t)dt}$ come from?*

Step 1: 兩邊乘 $\mu \rightarrow \mu\cdot dy/dt + \mu\cdot P\cdot y = \mu\cdot Q$

Step 1: *Multiply both sides by $\mu \rightarrow \mu\cdot dy/dt + \mu\cdot P\cdot y = \mu\cdot Q$*

Step 2: 希望左邊 $= d/dt[\mu\cdot y] = \mu\cdot dy/dt + d\mu/dt\cdot y$

Step 2: *Want LHS $= d/dt[\mu\cdot y] = \mu\cdot dy/dt + d\mu/dt\cdot y$*

Step 3: 比對 $\rightarrow d\mu/dt = \mu\cdot P$

Step 3: *Compare $\rightarrow d\mu/dt = \mu\cdot P$*

Step 4: 分離變數 $\rightarrow d\mu/\mu = P\,dt \rightarrow \ln \mu = \int P\,dt \rightarrow \mu = e^{\int P\,dt}$

Step 4: *Separate variables $\rightarrow d\mu/\mu = P\,dt \rightarrow \ln \mu = \int P\,dt \rightarrow \mu = e^{\int P\,dt}$*

Step 5: $d/dt[\mu\cdot y] = \mu\cdot Q \rightarrow$ 兩邊積分 $\rightarrow y = (1/\mu)(\int \mu\cdot Q\,dt + C)$

Step 5: *$d/dt[\mu\cdot y] = \mu\cdot Q \rightarrow$ Integrate both sides $\rightarrow y = (1/\mu)(\int \mu\cdot Q\,dt + C)$*

應用：$\tau\cdot dY/dt + Y = KpA \rightarrow P=1/\tau,\ \mu=e^{t}/\tau \rightarrow Y(t) = KpA(1-e^{-t}/\tau)$ ✓

Application: *$\tau\cdot dY/dt + Y = KpA \rightarrow P=1/\tau,\ \mu=e^{t}/\tau \rightarrow Y(t) = KpA(1-e^{-t}/\tau)$* ✓

練習題：解 $dy/dt + 2y = 6$，$y(0) = 0$。

Practice: *Solve $dy/dt + 2y = 6$, $y(0) = 0$.*

Step 1: $\mu = e^{\int 2\,dt} = e^{2t} \rightarrow d/dt(e^{2t}\,y) = 6e^{2t}$

Step 2: $e^{2t}\,y = 3e^{2t} + C \rightarrow y = 3 + Ce^{-2t}$

Step 3: $y(0) = 0 \rightarrow C = -3$

✓ $y(t) = 3(1 - e^{-2t})$

課程連結：Ch1-2（Unit 4）使用 $\mu=e^{t}/\tau$ 的完整來源。Ch4（溫度計一階推導）也需要積分因子。

Connection: *Ch1-2 (Unit 4) full derivation of $\mu=e^{t}/\tau$. Also needed in Ch4 (thermometer first-order derivation).*

M9. Taylor 展開與線性化 Taylor Series & Linearization

Taylor 展開：$f(x) = f(a) + f'(a)(x-a) + f''(a)(x-a)^2/2! + \cdots$

Taylor expansion: *$f(x) = f(a) + f'(a)(x-a) + f''(a)(x-a)^2/2! + \cdots$*

一階線性化： $f(x) \approx f(x_0) + f'(x_0)(x - x_0)$

First-order linearization: *$f(x) \approx f(x_0) + f'(x_0)(x - x_0)$*

直覺理解：在操作點附近，非線性函數近似為切線。偏離越遠，誤差越大。

Intuition: *Near any operating point, a smooth nonlinear curve looks like its tangent line. The further away, the larger the error.*

範例：線性化 $q = C\sqrt{h} \rightarrow q \approx q_s + (C/2\sqrt{h_s})(h - h_s) = q_s + (1/R_1)\cdot(h - h_s)$

Example: *Linearize $q = C\sqrt{h} \rightarrow q \approx q_s + (C/2\sqrt{h_s})(h - h_s) = q_s + (1/R_1)\cdot(h - h_s)$*

練習題：在 $x_0 = 4$ 處線性化 $f(x) = \sqrt{x}$。

Practice: *Linearize $f(x) = \sqrt{x}$ around $x_0 = 4$.*

Step 1: $f(4) = 2$, $f'(x) = 1/(2\sqrt{x})$, $f'(4) = 1/4$

✓ $\sqrt{x} \approx 2 + (1/4)(x - 4)$

課程連結：Ch5（液位槽 $q=C\sqrt{h}$ 線性化）、Ch7（Padé Taylor 匹配）、Ch12（案例分析中的線性化模型）。

Connection: *Ch5 (liquid-level $q=C\sqrt{h}$ linearization), Ch7 (Padé approximation via Taylor matching), Ch12 (linearized models in case studies).*

M10. 對數與分貝 Logarithms & Decibels

性質：$\log(ab) = \log a + \log b$；$\log(a/b) = \log a - \log b$；$\log(a^n) = n\cdot\log a$

Properties: *$\log(ab) = \log a + \log b$; $\log(a/b) = \log a - \log b$; $\log(a^n) = n\cdot\log a$*

分貝：$dB = 20 \log_{10}(AR)$ AR=10 → 20dB | AR=1 → 0dB | AR=0.707 → −3dB | AR=0.1 → −20dB

Decibels: *$dB = 20 \log_{10}(AR)$ AR=10 → 20dB | AR=1 → 0dB | AR=0.707 → −3dB | AR=0.1 → −20dB*

為何用 log：TF 相乘 → dB 相加；壓縮寬動態範圍，方便圖形化。

Why log scale: *Multiplying transfer functions = adding dB. Compresses wide dynamic range for plotting.*

練習題：將 AR = 0.5 和 AR = 10 轉換為 dB。

Practice: *Convert AR = 0.5 and AR = 10 to dB.*

Step 1: $dB = 20 \log_{10}(AR)$

Step 2: AR = 0.5: $20 \log_{10}(0.5) = 20\times(-0.301) = -6.02$ dB

✓ AR = 10: $20 \log_{10}(10) = 20\times1 = 20$ dB

課程連結：Ch4（Bode 圖介紹）、Ch15（Bode 建構）、Ch16（增益/相位裕度以 dB 表示）。

Connection: *Ch4 (Bode plot introduction), Ch15 (Bode diagram construction), Ch16 (gain/phase margin in dB).*

M11. 終值定理與初值定理 Final & Initial Value Theorems

終值定理 FVT：$\lim(t\to\infty)\ f(t) = \lim(s\to 0)\ sF(s)$（前提：$sF(s)$ 極點皆在 LHP）

Final Value Theorem: *$\lim(t\to\infty)\ f(t) = \lim(s\to 0)\ sF(s)$ (provided all poles of $sF(s)$ are in LHP)*

初值定理 IVT：$\lim(t\to 0^+)\ f(t) = \lim(s\to\infty)\ sF(s)$

Initial Value Theorem: *$\lim(t\to 0^+)\ f(t) = \lim(s\to\infty)\ sF(s)$*

範例 FVT：$Y(s) = KpA/[s(\tau s+1)] \to \lim(s\to 0)\ s\cdot Y(s) = KpA$ ✓

FVT Example: *$Y(s) = KpA/[s(\tau s+1)] \to \lim(s\to 0)\ s\cdot Y(s) = KpA$ ✓*

練習題：已知 $F(s) = 3/(s(s+2))$，求 $f(\infty)$ 和 $f(0^+)$。

Practice: *Given $F(s) = 3/(s(s+2))$, find $f(\infty)$ and $f(0^+)$.*

Step 1: FVT: $f(\infty) = \lim(s\to 0)\ s\cdot 3/(s(s+2)) = \lim(s\to 0)\ 3/(s+2) = 3/2$

✓ IVT: $f(0^+) = \lim(s\to\infty)\ s\cdot 3/(s(s+2)) = \lim(s\to\infty)\ 3/(s+2) = 0$

課程連結：Ch3（驗證反 Laplace 結果）、Ch8（穩態誤差 / offset）、Ch11（閉迴路穩態分析）、Ch12（offset 計算）。

Connection: *Ch3 (verify inverse Laplace), Ch8 (steady-state error / offset), Ch11 (closed-loop steady-state), Ch12 (offset calculations).*

M12. 多項式長除法 Polynomial Long Division

何時需要：分子次數 ≥ 分母次數（improper fraction）→ 先除再做 PFD。

When needed: *Numerator degree ≥ denominator degree (improper fraction) → divide first, then PFD.*

範例：$(s^2+3s+2)/(s+1) \to s^2\div s = s$，餘 $2s+2 \to 2s\div s = 2$，餘 0 → 結果 = $s+2$

Example: *$(s^2+3s+2)/(s+1) \to s^2\div s = s$, remainder $2s+2 \to 2s\div s = 2$, remainder 0 → result = $s+2$*

練習題：計算 $(s^2 + 3s + 5) \div (s + 1)$。

Practice: *Compute $(s^2 + 3s + 5) \div (s + 1)$.*

Step 1: $s^2 + 3s + 5 = (s + 1)\cdot(s + 2) + 3$

✓ $= s + 2 + 3/(s + 1)$

課程連結：Ch3（PFD 前處理，需先化為 proper fraction）、Ch6（高階 TF 化簡）。

Connection: *Ch3 (PFD requires proper fractions — divide first if improper), Ch6 (higher-order TF simplification).*

M13. 特徵方程式、極點與零點 Characteristic Equation, Poles & Zeros

定義：G(s) = N(s)/D(s)。極點 = D(s)=0 的根；零點 = N(s)=0 的根。

Definition: *G(s) = N(s)/D(s). Poles = roots of D(s)=0; Zeros = roots of N(s)=0.*

特徵方程式（閉迴路）：1 + G_OL(s) = 0

Characteristic equation (closed-loop): *1 + G_OL(s) = 0*

穩定性：所有極點 Re(s) < 0 → 穩定。 s = −a → e^{-at}（衰減）| s = −a±jb → 衰減振盪 | s = ±jb → 持續振盪 | s = +a → e^{at}（發散）

Stability: *All poles Re(s) < 0 → stable. s = −a → e^{-at} (decay) | s = −a±jb → damped oscillation | s = ±jb → sustained oscillation | s = +a → e^{at} (diverge)*

練習題：求 G(s) = (s+1)/((s+2)(s+3)) 的極點、零點，判斷穩定性。

Practice: *Find poles, zeros of G(s) = (s+1)/((s+2)(s+3)) and determine stability.*

Step 1: Zero: s = −1 (numerator = 0)

Step 2: Poles: s = −2, s = −3 (denominator = 0)

✓ All poles in LHP (Re < 0) → Stable

課程連結：Ch3（六種根型態）、Ch6（串聯極點）、Ch7（二階阻尼）、Ch8（閉迴路特徵方程式）、Ch11（穩定性分析）、Ch12（案例分析）、Ch13（BIBO 穩定性）、Ch17（根軌跡）。

Connection: *Ch3 (6 root types), Ch6 (cascaded poles), Ch7 (2nd-order damping), Ch8 (closed-loop char. eq.), Ch11 (stability), Ch12 (case studies), Ch13 (BIBO), Ch17 (root locus).*

M14. Routh-Hurwitz 穩定性判據 Routh-Hurwitz Stability Criterion

目的：不解多項式即可判斷根是否全在 LHP（左半平面）。

Purpose: *Determine whether all roots lie in the LHP without solving the polynomial.*

範例：$s^3 + 6s^2 + 11s + 6 = 0$ Row 1: 1, 11 | Row 2: 6, 6 | Row 3: (6·11−1·6)/6 = 10 | Row 4: 6 第一列 1, 6, 10, 6 全正 → 穩定（根 = −1, −2, −3）

Example: *$s^3 + 6s^2 + 11s + 6 = 0$ Row 1: 1, 11 | Row 2: 6, 6 | Row 3: (6·11−1·6)/6 = 10 | Row 4: 6 First column 1, 6, 10, 6 all positive → stable (roots = −1, −2, −3)*

判定規則：第一列符號變化次數 = RHP 根個數。0 次變化 → 穩定。

Criterion: *Number of sign changes in first column = number of RHP roots. Zero changes → stable.*

練習題：用 Routh 判據檢驗 $s^3 + 4s^2 + 5s + 2 = 0$ 是否穩定。

Practice: *Use Routh criterion to test stability of $s^3 + 4s^2 + 5s + 2 = 0$.*

Routh Array: s^3 | 1 5 s^2 | 4 2 s^1 | (4·5−1·2)/4 = 18/4 = 4.5 s^0 | 2

✓ First column: 1, 4, 4.5, 2 — all positive → Stable

課程連結：Ch11（閉迴路穩定性分析）、Ch12（案例分析中的穩定性）、Ch13（BIBO 穩定性與 Routh 測試）。

Connection: *Ch11 (closed-loop stability analysis), Ch12 (stability in case studies), Ch13 (BIBO stability & Routh test).*

M15. 頻率響應 s=jω Frequency Response

方法：G(s) 中令 $s = j\omega \rightarrow G(j\omega) = Re(\omega) + j\cdot Im(\omega)$

Method: *Substitute $s = j\omega$ into $G(s) \rightarrow G(j\omega) = Re(\omega) + j\cdot Im(\omega)$*

$AR = |G(j\omega)| = \sqrt{Re^2 + Im^2}$ $\varphi = \arctan(Im/Re)$

範例：$G(s) = 1/(\tau s+1) \rightarrow G(j\omega) = (1-j\omega\tau)/(1+\omega^2\tau^2)$ $AR = 1/\sqrt{1+\omega^2\tau^2}$, $\varphi = -\arctan(\omega\tau)$

$\omega=1/\tau$ 時：$AR = 1/\sqrt{2} \approx 0.707$ (−3dB), $\varphi = -45°$

Example: *$G(s) = 1/(\tau s+1) \rightarrow G(j\omega) = (1-j\omega\tau)/(1+\omega^2\tau^2)$ $AR = 1/\sqrt{1+\omega^2\tau^2}$, $\varphi = -\arctan(\omega\tau)$ At $\omega=1/\tau$: $AR = 1/\sqrt{2} \approx 0.707$ (−3dB), $\varphi = -45°$*

練習題：已知 $G(s) = 2/(s+1)$，求 $\omega = 1$ 時的 $|G(j\omega)|$ 和 $\angle G(j\omega)$。

Practice: *Given $G(s) = 2/(s+1)$, find $|G(j\omega)|$ and $\angle G(j\omega)$ at $\omega = 1$.*

Step 1: $G(j1) = 2/(1+j1) = 2/(1+j)$

Step 2: $|G| = 2/\sqrt{1^2+1^2} = 2/\sqrt{2} = \sqrt{2} \approx 1.414$

✓ $\angle G = -\arctan(1/1) = -45°$

課程連結：Ch4（一階 Bode 圖）、Ch7（二階共振）、Ch15（Bode 圖建構）、Ch16（增益/相位裕度設計）。

Connection: *Ch4 (first-order Bode), Ch7 (second-order resonance), Ch15 (Bode diagram construction), Ch16 (gain/phase margin design).*

M16. Padé 近似 Padé Approximation

目的：用有理函數近似 dead time $e^{-\theta s}$，使其可用於 Laplace 域分析。

Purpose: *Approximate dead time $e^{-\theta s}$ with a rational function for Laplace-domain analysis.*

一階 Padé / First-order Padé: $e^{-\theta s} \approx (1 - \theta s/2)/(1 + \theta s/2)$

來源：在 s=0 匹配 Taylor 展開到 $O(s^2)$。 AR = 1（全通，與真實 dead time 相同）。RHP 零點 $s = 2/\theta \rightarrow$ 可能造成 inverse response。

Origin: *Matches Taylor expansion at s=0 up to $O(s^2)$. AR = 1 (all-pass, same as true dead time). RHP zero at $s = 2/\theta \rightarrow$ may cause inverse response.*

練習題：求 e^{-s} 的 [1/1] Padé 近似。

Practice: *Find the [1/1] Padé approximation of e^{-s}.*

Step 1: $e^{-s} \approx 1 - s + s^2/2 - \cdots$, 設 $\approx (1 + as)/(1 + bs)$

Step 2: Match: $a = -1/2$, $b = 1/2$

✓ $e^{-s} \approx (1 - s/2)/(1 + s/2) = (2 - s)/(2 + s)$

課程連結：Ch7（dead time / 傳輸延遲近似）、Ch12（含 dead time 的案例分析）。

Connection: *Ch7 (dead time / transportation lag approximation), Ch12 (case study with dead time and Padé).*